

$$\frac{d}{dx} \sin(x) = \cos(x) \quad \frac{d}{dx} \cos(x) = -\sin(x) \quad \frac{d}{dx} \tan(x) = \sec^2(x) \quad \frac{d}{dx} \sec(x) = \sec(x) \tan(x)$$

$$\frac{d}{dx} \csc(x) = -\csc(x) \cot(x) \quad \frac{d}{dx} \cot(x) = -\csc^2(x)$$

$$\int \cos(x) dx = \sin(x) + C \quad \int \sin(x) dx = -\cos(x) + C \quad \int \sec^2(x) dx = \tan(x) + C$$

$$\int \sec(x) \tan(x) dx = \sec(x) + C \quad \int \csc(x) \cot(x) dx = -\csc(x) + C \quad \int \csc^2(x) dx = -\cot(x) + C$$

$$\frac{d}{dx} \sin^{-1}(x) = \frac{1}{\sqrt{1-x^2}} \quad \frac{d}{dx} \cos^{-1}(x) = \frac{(-1)}{\sqrt{1-x^2}} \quad \frac{d}{dx} \tan^{-1}(x) = \frac{1}{(1+x^2)} \quad \frac{d}{dx} \cot^{-1}(x) = \frac{(-1)}{(1+x^2)}$$

$$\frac{d}{dx} \sec^{-1}(x) = \frac{1}{x\sqrt{x^2-1}} \quad \frac{d}{dx} \csc^{-1}(x) = \frac{(-1)}{x\sqrt{x^2-1}}$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C \quad \int \frac{1}{\sqrt{1+x^2}} dx = \tan^{-1} x + C \quad \int \frac{1}{x} \sqrt{x^2-1} dx = \sec^{-1} x + C$$

$$b^{-(n)} = \frac{1}{b^n} \quad \ln(x^n) = n \ln(x) \quad \int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + c \quad \int u dv = uv - \int v du \quad \ln^n(x) = (\ln x)^n$$

$$\mathcal{L}\{f(t)\} = \int_0^\infty f(t)e^{-st} dt$$

$$\mathcal{L}\{c\} = \frac{\{c\}}{s} \quad \mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}} \quad \mathcal{L}\{e^{\{c\}t}\} = \frac{1}{s-\{c\}} \quad \mathcal{L}\{\sin(kt)\} = \frac{k}{s^2+k^2} \quad \mathcal{L}\{\cos(kt)\} = \frac{s}{s^2+k^2}$$

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$$

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$$

$$\mathcal{L}\{f''(t)\} = s^2 F(s) - sf(0) - f'(0) \quad \mathcal{L}\{f'''(t)\} = s^3 F(s) - s^2 f(0) - sf'(0) - f''(0)$$

$$\sin(x) \cos(x) = \frac{1}{2} \sin(2x) \quad \sin^2(x) = \frac{1-\cos^2(x)}{2} \quad \cos^2(x) = \frac{1+\cos^2(x)}{2}$$

other stuff

$$\begin{array}{ccc} y=12-6x & y=12-6x & x=\frac{-1}{6}y+2 \\ \int_0 & \int_0 & \int_0 \end{array} \quad \begin{array}{l} dy \text{ is correct} \\ dx \text{ is incorrect (integration order change)} \\ dx \text{ is now correct} \end{array}$$

for determining the bounds of integration:

its like thinking of either side of as a scanner, in this case, x is our scanner line, as we set the bounds from 0 to 5 there is a veritcal line that scans the whatever lines are on the point, from x = 0 to x = 5, which in this case we have two lines x and -x + 10. dx scans every dy. so outer bounds is the scanner and the inner bounds is what is(the lines, equations) scanned and the area is taken by the distance between those equations

R: start length $\leq r \leq$ end length, given the angles, start space $\leq \theta \leq$ end space

triple integral stuff:

if you see x^2 and y^2 think of $x^2 + y^2 = r^2$ (circle) or $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (ellipse)

cylindrical coordinates $(x, y, z) = (r, \theta, z)$

$$x = r \cos \theta \quad y = r \sin \theta \quad r = \sqrt{x^2 + y^2} \quad \theta = \tan^{-1}\left(\frac{y}{x}\right) \quad z = z \quad r^2 = x^2 + y^2$$

spherical coordinates $(x, y, z) = (\rho, \theta, \varphi)$ $\rho \geq 0$ $0 \leq \varphi \leq \pi$

$$r = \rho \sin(\varphi) \quad \theta = \theta \quad z = \rho \cos(\varphi)$$

$$\rho^2 = z^2 + r^2 \quad p^2 = x^2 + y^2 + z^2$$

$$x = \rho \sin(\varphi) \cos(\theta) \quad y = \rho \sin(\varphi) \sin(\theta) \quad z = \rho \cos(\varphi)$$

use jacobian when switching coord. sys.

in circ. polar, factor = r

in ellipt. polar factor = $ab \cdot r$

the order of integration is usually $dr d\theta$ (from $dx dy$)

boundries: cart. = -r to r polar = 0 to r

to find centroid, find the integral of the shape, this will be M redo integral but start with x, this will be M_y this will give you the x coordinate by $\bar{x} = \frac{M_y}{M}$ redo again but with y, this will give you M_x this will give you the y coordinate by $\bar{y} = \frac{M_x}{M}$

average height of a function: $\frac{1}{A} \int \int f(x, y) dx dy$

$$\int_a^b \int_c^d f(x, y) dx dy$$

jacobian 2d:

$$J(u, v) = \frac{\partial y}{\partial v} \frac{\partial x}{\partial u} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v} \text{ (solve the deriv of denom. wrt num.)}$$

cylindrical coord.

$$J = r$$

spherical coord.

$$J = \rho^2 \sin(\varphi)$$

jacob sub.

find y of u by plugging in x of v into u

find x of v by plugging y of u into v

then do jacob.

plug in y of u and x of v into the given equations those are the bounds. order of integration is $du dv$